DGMS 2019 • SECTION D • QUESTION 27

Equilibrium Mineral Assemblages: The Forsterite–Quartz Incompatibility & Phase Petrology

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Igneous Petrology & Phase Equilibria Guide

Question Statement

Q.27. Which of the following do NOT constitute an equilibrium mineral assemblage in a rock?

a (a) Quartz and orthoclase

b **(b) Forsterite and quartz**

c (c) Hornblende and andesine

d (d) Labradorite and augite

Correct Answer: (b) Forsterite and quartz (Chemically incompatible pair)

Official Textbook Phase Diagrams (Winter 2010, Chapter 7)

derstand it, we can then add other components to approach more realistic magmatic systems. What does this simplified Di-An basalt system tell us? It explains once again how liquids of more than one component crystallize over a *range* of temperatures, even without solid solution (observation 1 concerning the behavior of natural melts in Section 1). Second, it explains how we can get a *sequence* of minerals crystallizing as a basalt cools and that the sequence varies with composition (observations 2, 3, and 7). If the composition is to the right of the eutectic, anorthite will form, followed by diopside (+ anorthite). In ophitic texture in a basalt, earlier euhedral to subhedral plagioclase crystals are surrounded by later interstitial augite. If the bulk composition is to the left of the eutectic, augite crystals form first. Also note that the Di-An system suggests that the composition of initial partial melts, even of more complex natural plagioclase– clinopyroxene rocks, ought to be concentrated around a specific composition (the eutectic).

Figure 11 illustrates the common tendency for an added component to lower the melting point of the complementary one-component system. Adding a diopside component to pure anorthite or an anorthite component to pure diopside results in a lower melting point of the mixture than in the nearby pure system. In contrast to the plagioclase or olivine systems, the second component here results in a second mineral, not a single mineral with a variable composition. We can thus distinguish two different types of additional components: those that mix into the phase with the original component(s) and those that don't and thus require a new phase.

If we were to use thermodynamics to calculate the conditions under which this reaction takes place, we would find that under all realistic conditions, the reaction runs to the right, as written. In other words, *Mg-rich olivine and quartz can never coexist in equilibrium in igneous rocks*! If together, they would react to form orthopyroxene until one or the other were consumed. As a result, there is no field labeled "forsterite + quartz" (or any silica polymorph) anywhere in

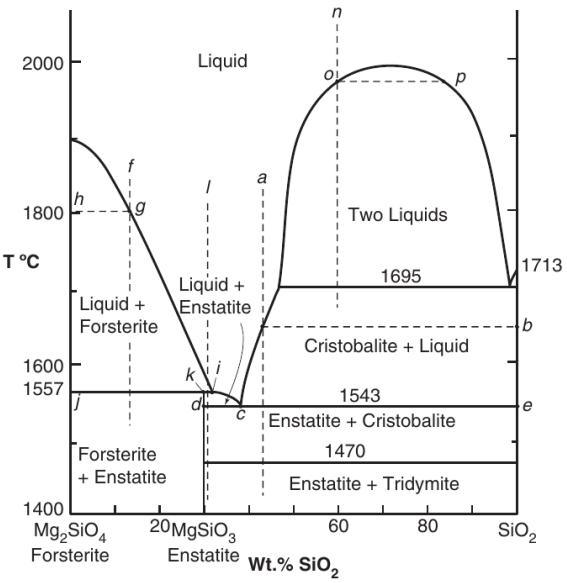


FIGURE 12 Isobaric *T-X* phase diagram of the system Fo-silica at 0.1 MPa. After Bowen and Anderson (1914) and Grieg (1927). Reprinted by permission of the *American Journal of Science*.

Figure 1: John D. Winter (2010), Chapter 7, Figure 12. Isobaric T–X phase diagram of the binary system **Forsterite–Silica (Mg_2SiO_4–SiO_2)** at 0.1 MPa (1 atm). Note the intermediate compound **Enstatite (En: $MgSiO_3$)**, which melts incongruently at 1557°C to Forsterite + Liquid. At equilibrium, the system is split into two mutually exclusive sub-systems: **(1) Fo + En** (silica-undersaturated) and **(2) En + Qtz** (silica-oversaturated). **Forsterite and Quartz can never coexist across the intermediate Enstatite phase barrier!**

one of the reactants is consumed. This is an important point but is often overlooked. We tend to look at reactions such as Reaction (10) and think that both reactants are completely consumed as the product is produced. *This will really happen only in the rare circumstances that the reactants exist in exactly the correct proportions so that they are exhausted simultaneously*. One will typically be exhausted first. The other, with no remaining co-conspirator with which to react, remains perfectly stable in the system with the product(s). In the present situation, which reactant is consumed first by the reaction? We can answer this question geometrically on Figure 12 by applying the lever principle in a slightly different way. Figure 13a illustrates the geometry of the peritectic tie-line. I have shifted the positions of the phases from Figure 12 a little to aid visualization and used mineral abbreviations to save space. Reaction (10) involves the two line-end phases (Fo + Liq) combining to form the phase between them (En). The lever principle tells us that Fo and Liq must be combined (react) in the proportions Fo:Liq $= \overline{kj} : \overline{jk}$ in order to produce En. This is reasonable because En plots closer to Liq than to Fo, so it must contain less Fo than it does liquid components. If the bulk composition of our melt were exactly equal to En, then Fo and Liq would both be exhausted at the same instant. If the bulk composition plots between Fo and En, however (the gray dot in Figure 13b), there is an excess of Fo, and the liquid would be consumed first, leaving forsterite and enstatite. Alternatively, if the bulk composition plots between En and Liq (the gray dot in Figure 13c), there would be an excess of liquid, and forsterite would be consumed first.

In the present situation, the bulk composition (f) lies between enstatite and forsterite, so the liquid must be consumed first. Once the liquid is used up, the remaining forsterite will coexist with enstatite, $\phi \rightarrow 2$, $F \rightarrow 1$, and the

Figure 2: John D. Winter (2010), Chapter 7, Figure 13. Schematic enlargement of the peritectic reaction tie-line: **Forsterite + Liquid (silica-rich)** ⇌ **Enstatite**. Any cooling magma containing excess SiO_2 consumes all forsterite to form enstatite before quartz can ever crystallize.

toward quartz. There are no fancy shifts across the enstatite composition, as in the peritectic system. At these high pressures, it becomes impossible to extract a silica-rich melt from which cristobalite can crystallize from an original mixture of olivine and enstatite. Also, at high pressures, there should be no peritectic reaction upon cooling between olivine and liquid to produce enstatite, so embayed olivine crystals, or orthopyroxene rims on olivine, should be absent.

5.4 The Alkali Feldspar System

As a final example of binary systems, we address the $NaAlSi_3O_8$-$KAlSi_3O_8$ system (Ab-Kfs, or the alkali feldspar system). The somewhat simplified *T*-*X* phase diagram (at a

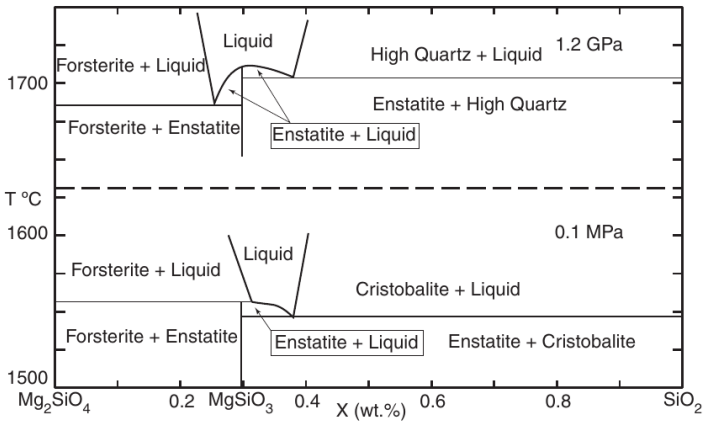


FIGURE 15 The system Fo-SiO_2 at atmospheric pressure and 1.2 GPa. After Bowen and Schairer (1935), Chen and Presnall (1975). Copyright © Mineralogical Society of America.

a single feldspar at lower temperatures as $\psi \rightarrow 1$ and $\ldots \rightarrow 2$ in the divariant field labeled "single feldspar" in Figure 16. With a single feldspar, we must define *T* and X_{Ab}^{feld} or X_{Or}^{feld} to determine the system.

Cooling a melt of composition *i* in Figure 16 would result in crystallization of a Na-rich feldspar of composition *j* at about 1000°C, coexisting with a more K-rich melt. Cooling would cause both the liquid and feldspar to become *less* sodic as the compositions follow the liquidus and solidus, respectively. The final liquid would have the composition *k*, and a single feldspar of composition *i* would exist below this.

Because there is solid solution, **fractional crystallization** would affect this system, shifting the last liquid (and solid) compositions closer to the eutectic point, regardless of which side of the eutectic the bulk composition lies.

Figure 3: John D. Winter (2010), Chapter 7, Figure 15. The Fo–SiO_2 system at atmospheric pressure (bottom) and elevated pressure (top). At elevated pressures, enstatite melts congruently, creating a permanent **thermal divide** between Fo-bearing rocks and Qtz-bearing rocks.

In-Depth Option Evaluation & Petrographic Analysis

(b) Forsterite and quartz — [DOES NOT CONSTITUTE AN EQUILIBRIUM ASSEMBLAGE • CORRECT]

Chemical Incompatibility: Forsterite (Mg_2SiO_4) is a highly silica-undersaturated nesosilicate (olivine group). Quartz (SiO_2) is pure silica (tectosilicate). In the MgO–SiO_2 system, an intermediate chain silicate phase exists: **Enstatite ($MgSiO_3$)**.

The Incompatibility Reaction: Under all geological crustal and upper-mantle conditions (T < 1557°C and P < 10 GPa), forsterite and quartz react completely and irreversibly to form enstatite:

Forsterite–Quartz Breakdown Reaction:

Mg_2SiO_4 (Forsterite) + SiO_2 (Quartz) → 2 $MgSiO_3$ (Enstatite)

$\Delta G^\circ_{reaction} < 0$ (Thermodynamically spontaneous)

Because the Gibbs free energy of this reaction is strongly negative, a rock can contain either **(Forsterite + Enstatite)** in silica-undersaturated compositions, or **(Enstatite + Quartz)** in silica-oversaturated compositions, but **NEVER Forsterite + Quartz** together in thermodynamic equilibrium!

Note on Fe-rich systems: While Mg-olivine (Forsterite) is incompatible with quartz, Fe-rich olivine (**Fayalite, Fe_2SiO_4**) CAN coexist with quartz in highly differentiated ferroan granites and rhyolites because pure ferrosilite ($FeSiO_3$) is unstable at low pressures. The question explicitly specifies **Forsterite**, making this an absolute thermodynamic incompatibility!

(a) Quartz and orthoclase — [STABLE EQUILIBRIUM PAIR]

Rock Types: Granite, Rhyolite, Granitic Pegmatite, Aplite, Felsic Gneiss.

Phase Relations: Quartz (SiO_2) and Orthoclase ($KAlSi_3O_8$) form the classic low-temperature eutectic in "Petrogeny's Residua System" (Quartz–Albite–Orthoclase). They crystallize together in mutual thermodynamic equilibrium as the terminal assemblage of granitic melts.

(c) Hornblende and andesine — [STABLE EQUILIBRIUM PAIR]

Rock Types: Diorite, Andesite, Amphibolite (metabasite).

Phase Relations: Hornblende (calcic amphibole) and Andesine (intermediate plagioclase, An_{30}–An_{50}) define the diagnostic mineral assemblage of intermediate calc-alkaline magmas as well as the standard **amphibolite facies** of regional metamorphism. They are in complete thermodynamic equilibrium.

(d) Labradorite and augite — [STABLE EQUILIBRIUM PAIR]

Rock Types: Basalt, Gabbro, Diabase / Dolerite.

Phase Relations: Labradorite (calcic plagioclase, An_{50}–An_{70}) and Augite (calcic clinopyroxene) form the principal cotectic mineral pair of mafic igneous rocks worldwide, coexisting stably in equilibrium to produce ophitic, subophitic, and intergranular textures.

Petrological Foundations: Silica Saturation & Bowen's Reaction Series

1. The Concept of Silica Saturation (Shand, 1927)

Igneous rocks and mineral assemblages are divided into three fundamental categories based on their silica activity:

Category	Definition	Diagnostic Minerals	Forbidden Minerals (Incompatible)
Silica-Oversaturated	Contains free primary SiO_2	Quartz, Cristobalite, Tridymite, Hypersthene	**Forsterite, Nepheline, Leucite, Sodalite**
Silica-Saturated	Exact stoichiometric silica; no free quartz, no unsaturated phases	Enstatite, Diopside, Feldspars (Albite, Anorthite, Orthoclase)	Quartz + Forsterite together
Silica-Undersaturated	Deficient in SiO_2; forms silica-poor minerals	Forsterite, Nepheline, Leucite, Kalsilite, Melilite	**Quartz**

2. Bowen's Discontinuous Reaction Series

N. L. Bowen (1928) demonstrated that during fractional crystallization of basaltic magmas:

Bowen's Discontinuous Branch:

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Olivine (Fo) → Orthopyroxene (En) → Clinopyroxene (Augite) → Amphibole (Hornblende) → Biotite

• At the peritectic reaction: Forsterite + Liquid → Enstatite.
• In a silica-rich melt, all forsterite is resorbed before quartz can form.
• In a silica-poor melt, quartz never forms because all silica is consumed.
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Direct Textbook Quotations & Citations

John D. Winter (2010), *Principles of Igneous and Metamorphic Petrology, Chapter 7, p. 121:*

"Olivine crystals begin to form at 1800°C, they continue to grow with cooling, and then they begin to be consumed back into the melt as a new mineral, **enstatite***, forms... You might also recognize this* **olivine → pyroxene transition as the first step in the 'discontinuous' left side of Bowen's Reaction Series***... Because enstatite forms between forsterite and silica,* **forsterite and quartz cannot coexist in equilibrium***."*

— Winter (2010), Chapter 7, p. 121

Anthony R. Philpotts & Jay J. Ague (2009), *Principles of Igneous and Metamorphic Petrology, 2nd ed., Chapter 10:*

"Because enstatite is intermediate in composition between forsterite and quartz, the reaction **Fo + Qtz = 2 En** *prevents the coexistence of forsterite and quartz at equilibrium in any igneous or metamorphic rock. Rocks containing olivine and quartz are either non-equilibrium mixtures or contain iron-rich olivine (fayalite)."*

— Philpotts & Ague (2009), Chapter 10

Exam Traps & Rapid Elimination Strategy

Standard Competitive Exam Incompatible Pairs (Forbidden Assemblages):

1. **Forsterite + Quartz:** Always incompatible! They react to form Enstatite / Orthopyroxene (*Fo + Qtz → 2 En*).
2. **Nepheline + Quartz:** Always incompatible! They react to form Albite (*Ne + 2 Qtz → Ab*).
3. **Leucite + Quartz:** Always incompatible! They react to form K-feldspar (*Lc + Qtz → Kfs*).
4. **Kalsilite + Quartz:** Always incompatible! They react to form K-feldspar (*Kls + 2 Qtz → Kfs*).
5. **Corundum + Quartz:** Always incompatible! They react to form Al_2SiO_5 (*Crn + Qtz → Ky/Sil/And*).